CLASS 8 — MATHS · FORMULA SHEET

Chapter 8: **Fractions in Disguise** | Quick revision, one page

Percentages

- $x\% = \frac{x}{100}$; $x\%$ of $N = \frac{x}{100} \times N$.
- Percentage change $= \frac{\text{change}}{\text{original}} \times 100$.
- After an increase of $x\%$: $N\left(1 + \frac{x}{100}\right)$; after a decrease: $N\left(1 - \frac{x}{100}\right)$.

Profit, loss, discount

- Profit $=$ SP $-$ CP (if SP $>$ CP); Loss $=$ CP $-$ SP (if CP $>$ SP).
- Profit% $= \frac{\text{Profit}}{\text{CP}} \times 100$; Loss% $= \frac{\text{Loss}}{\text{CP}} \times 100$.
- Discount $=$ MP $-$ SP; Discount% $= \frac{\text{Discount}}{\text{MP}} \times 100$.
- GST: price with GST $=$ price before tax $\times \left(1 + \frac{\text{rate}}{100}\right)$.

Simple interest

- $SI = \frac{P \times R \times T}{100}$; Amount $A = P + SI$.

Compound interest

- $A = P\left(1 + \frac{R}{100}\right)^n$ (interest compounded yearly); $CI = A - P$.
- For 1 year, CI = SI. For 2 years, CI $-$ SI $= P\left(\frac{R}{100}\right)^2$.
- Successive percentage changes: multiply the factors, e.g. $+10\%$ then -10% gives $1.1 \times 0.9 = 0.99$ (a 1% decrease).

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